

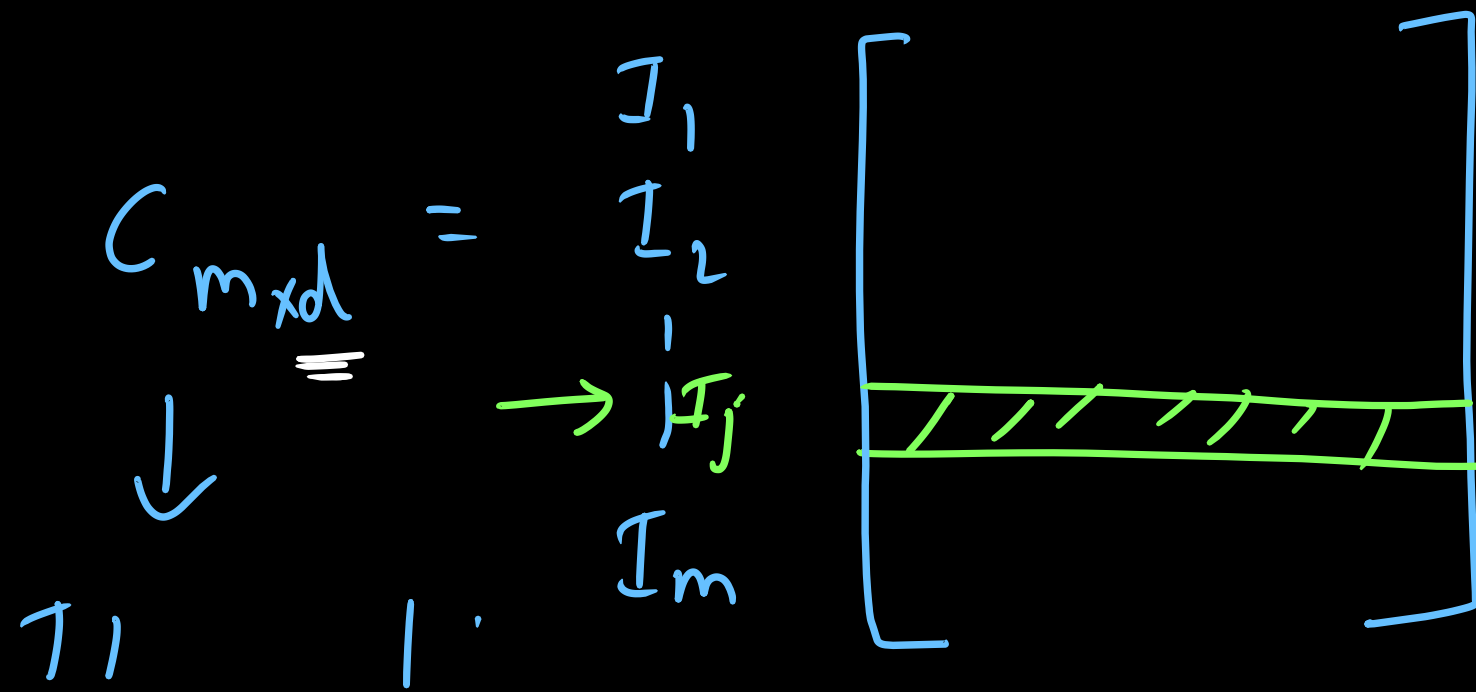
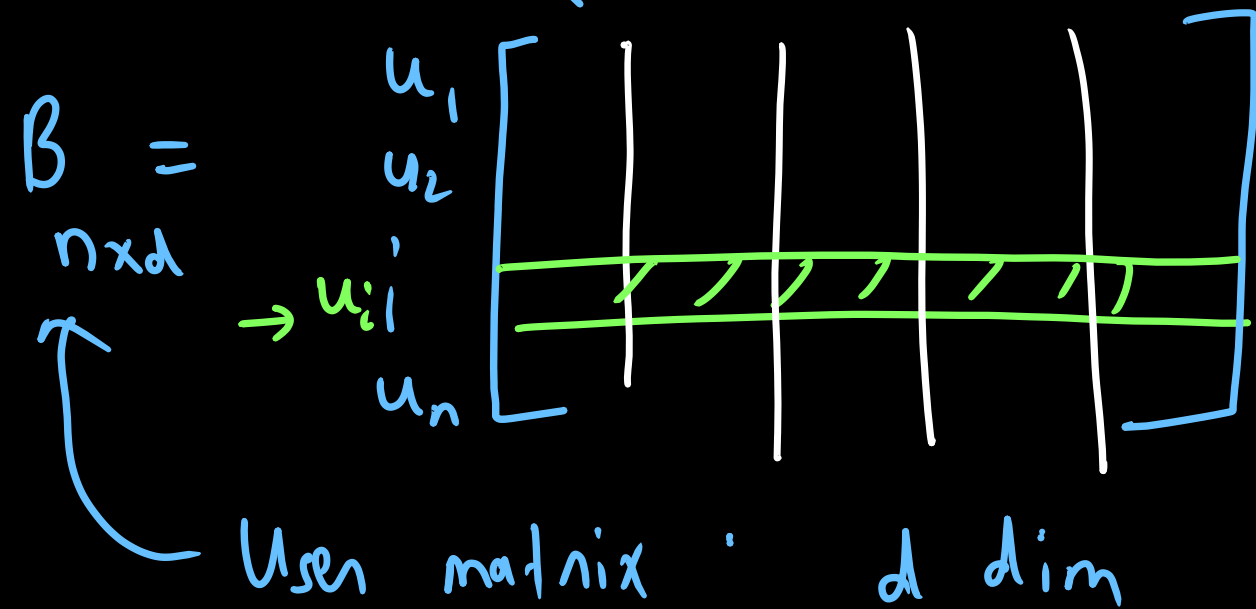
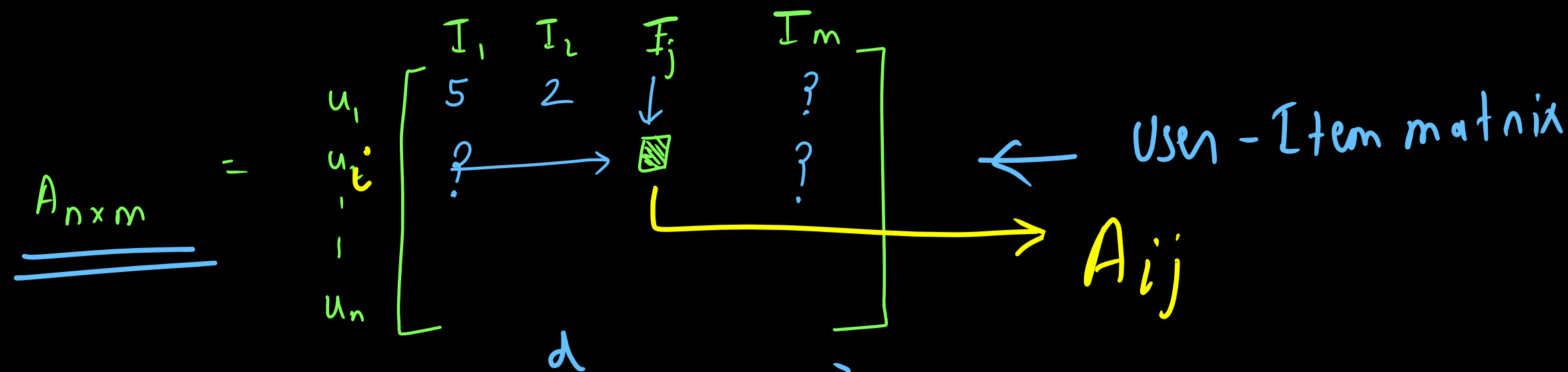
Agenda

- Related to K-means Clustering
- Relation to GMM
- Hyperparameter 'd'
- NMF

$$A_{n \times m} \approx B_{n \times d} C_{d \times m}^T$$

$$A_{ij} = B_i C_j^T$$

$| \times |$ $| \times d$ $d \times |$



- em mar /

$$B = \begin{matrix} & \begin{matrix} \leftarrow d \rightarrow \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{matrix} & \begin{bmatrix} 2.98 & 3 \\ 2.99 & 3.1 \\ 1 & 4 \end{bmatrix} \end{matrix}$$

$$\cos \theta = \frac{u_1 \cdot u_2}{\|u_1\| \|u_2\|} = 1$$

$$\theta \approx 0 \leftarrow \cosine(u_1, u_2) \rightarrow \text{high}$$

$$\cosine(u_1, u_3) \rightarrow \text{low}$$

$$\cos \theta = 0 \rightarrow \theta = 90^\circ$$

$u_1 \rightarrow$

Titanic, fault in our star, Scifi

$u_2 \rightarrow$

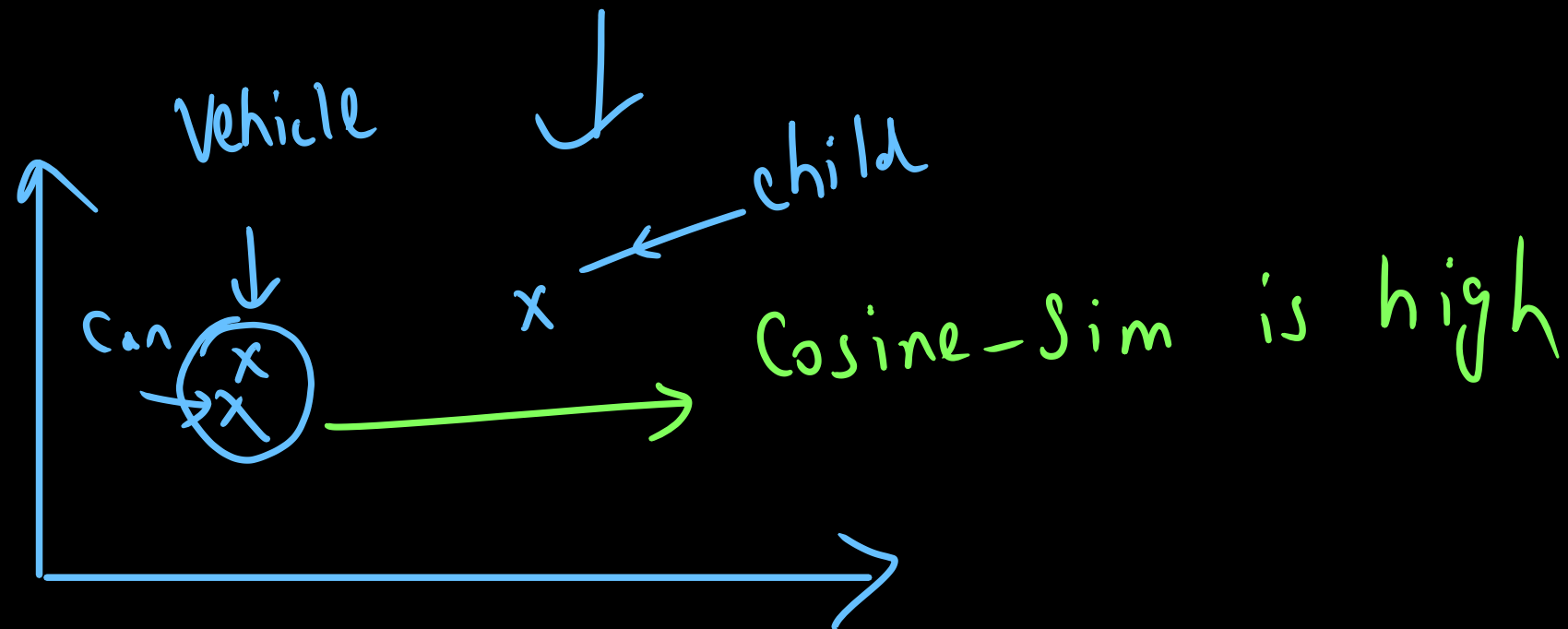
Titanic, fault in our star, Scifi

$u_3 \rightarrow$

Inception, Transformers, Prestige

1st → The Car is parked outside
2nd → The vehicle is parked outside
3rd → A child goes to school

Word2Vec



$$L = \min_{B_i, c_j} \sum_{A_{ij} \neq 0} (A_{ij} - B_i c_j^T)^2$$

$$= \min \|A - B c^T\|_F^2$$

$$\|A\|_F^2 = \sum \sum a_{ij}^2$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = 1^2 + 2^2 + 3^2 + 4^2$$

A: User - Item rating matrix

B: User matrix in d-dimension.

C: Item matrix in d-dimension.

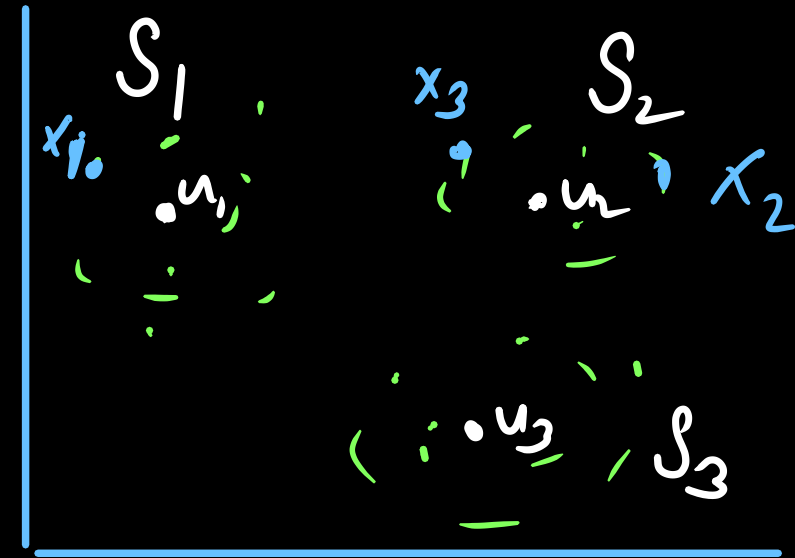
$c_1, c_2 \dots c_K$ are the cluster centroids ✓
 $S_1, S_2 \dots S_K$ represent the clusters

WCSS

$$L = \sum_{j=1}^K \sum_{\substack{x_i \in S_j}} \|x_i - c_j\|^2$$

↑
For each cluster

↑
For each datapoint in the cluster



$$z_{11}=1, \quad z_{12}=0, \quad z_{13}=0$$

$$z_{22}=1, \quad z_{32}=1$$

$$= \sum_{j=1}^K \sum_{i=1}^m z_{ij} \|x_i - c_j\|^2 \rightarrow \underbrace{z_{ij}=1}_{\checkmark} \rightarrow x_i \in S_j$$

K-means

✓
$$L = \min_{j=1}^K \sum_{x_i \in C_j} \|x_i - c_j\|^2$$

c_j was a vector

✓
$$L = \min_{C, Z} \|X - ZC^T\|_F^2$$

Cluster Assignment Matrix

Cluster centroid matrix

$$L = \min_{C, Z} \|X - ZC^T\|_F^2$$

$\swarrow$ $\searrow$
 $\underbrace{m \times d}_{\text{Cluster Assignment Matrix}}$ $\text{Cluster centroid matrix}$

$$z_{ij} = 1 \rightarrow x_i \in S_j$$

$$Z = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & \dots & K \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix} \end{matrix}$$

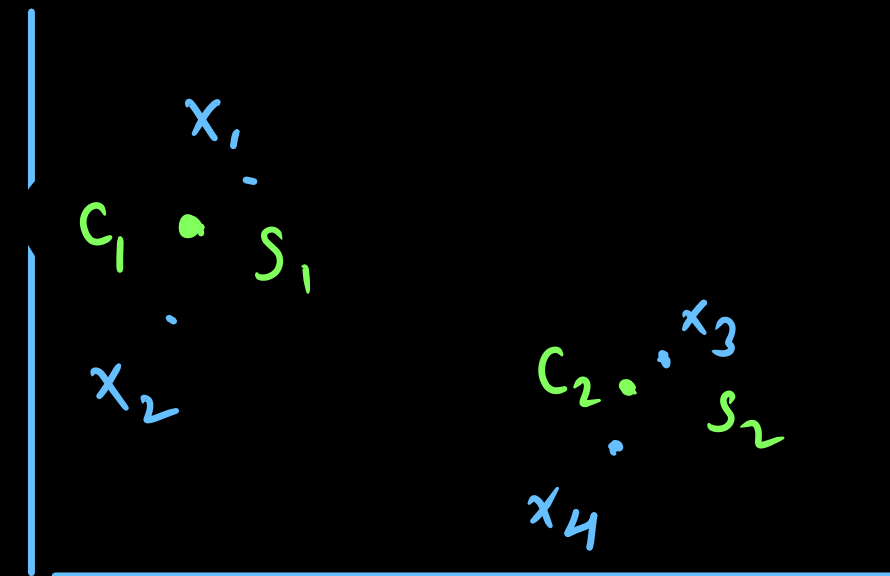
$\downarrow m$ $\rightarrow$ No of clusters
 No of data points

$$C = \begin{matrix} & \begin{matrix} \xleftarrow{K} \end{matrix} \\ \begin{matrix} \uparrow \\ d \\ \downarrow \end{matrix} & \begin{bmatrix} c_1 & c_2 & \dots & c_K \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \end{matrix}$$

$C = d \times K$

$X = 4 \times 3$

x_1	1	2	3	$\rightarrow S_1$
x_2	3	4	5	$\rightarrow S_1$
x_3	20	21	22	$\rightarrow S_2$
x_4	22	23	24	$\rightarrow S_2$



$\rightarrow c_1 \rightarrow (2, 3, 4)$ for cluster S_1
 $\rightarrow c_2 \rightarrow (21, 22, 23)$ for cluster S_2

$$L = \sum_{j=1}^K \sum_{x_i \in S_j} \|x_i - c_j\|^2$$

$$= \frac{(1-2)^2 + (2-3)^2 + (3-4)^2}{+ (3-2)^2 + (4-3)^2 + (5-4)^2}$$

$$+ (20-21)^2 + (21-22)^2 + (22-23)^2$$

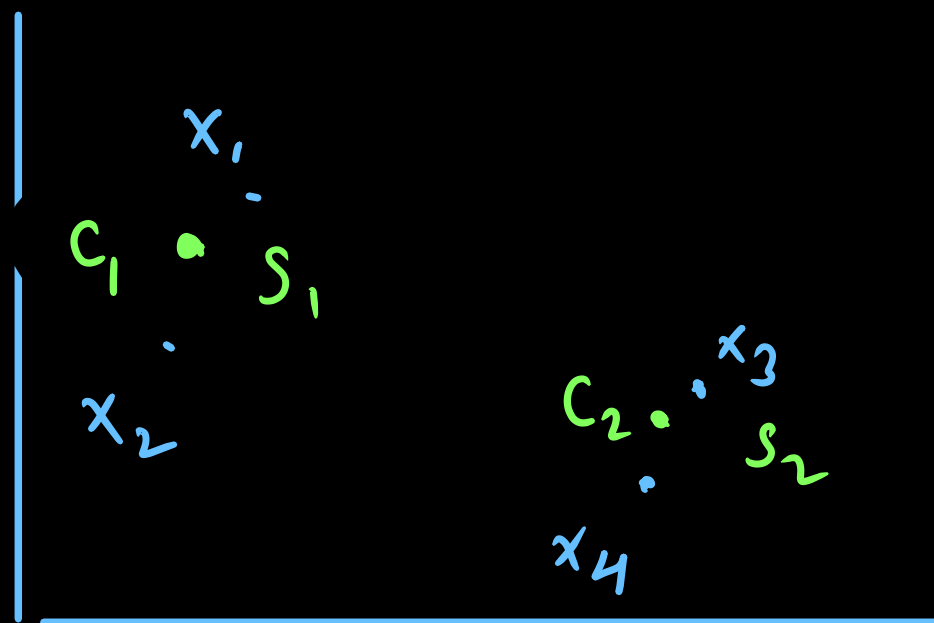
$$+ (22-21)^2 + (23-22)^2 + (24-23)^2$$

$$X = \begin{matrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 20 & 21 & 22 \\ 22 & 23 & 24 \end{bmatrix} \begin{matrix} \rightarrow s_1 \\ \rightarrow s_1 \\ \rightarrow s_2 \\ \rightarrow s_2 \end{matrix}$$

$$Z = \begin{matrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} \begin{bmatrix} & s_1 & s_2 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$C^T = \begin{bmatrix} \leftarrow c_1 \rightarrow \\ \leftarrow c_2 \rightarrow \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 21 & 22 & 23 \end{bmatrix}$$

$$ZC^T = \begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 4 \\ 21 & 22 & 23 \\ 21 & 22 & 23 \end{bmatrix}$$



$$\|X - ZC^T\|_F^2 = \left(\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 20 & 21 & 22 \\ 22 & 23 & 24 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 4 \\ 2 & 3 & 4 \\ 21 & 22 & 23 \\ 21 & 22 & 23 \end{bmatrix} \right)_F^2$$

$$= \begin{bmatrix} (1-2) & (2-3) & (3-4) \\ (3-2) & (4-3) & (5-4) \\ (20-21) & (21-22) & (22-23) \\ (22-21) & (23-22) & (24-23) \end{bmatrix}_F^2$$

$$= (1-2)^2 + (2-3)^2 + (3-4)^2 \\ + (3-2)^2 + (4-3)^2 + (5-4)^2 \\ - - - -$$

$$L = \min_{B_i, c_j} \sum_{A_{ij} \neq 0} (A_{ij} - B_i c_j')^2$$

$$= \min_{B, C} \|A - B C^T\|_F^2$$

— ① MF

$$L = \min_{C, Z} \|X - Z C^T\|_F^2$$

— ② K-means

Cluster
Assignment
Matrix

Cluster
centroid
matrix

GMM

→ Soft Cluster

→ A point can belong to multiple clusters

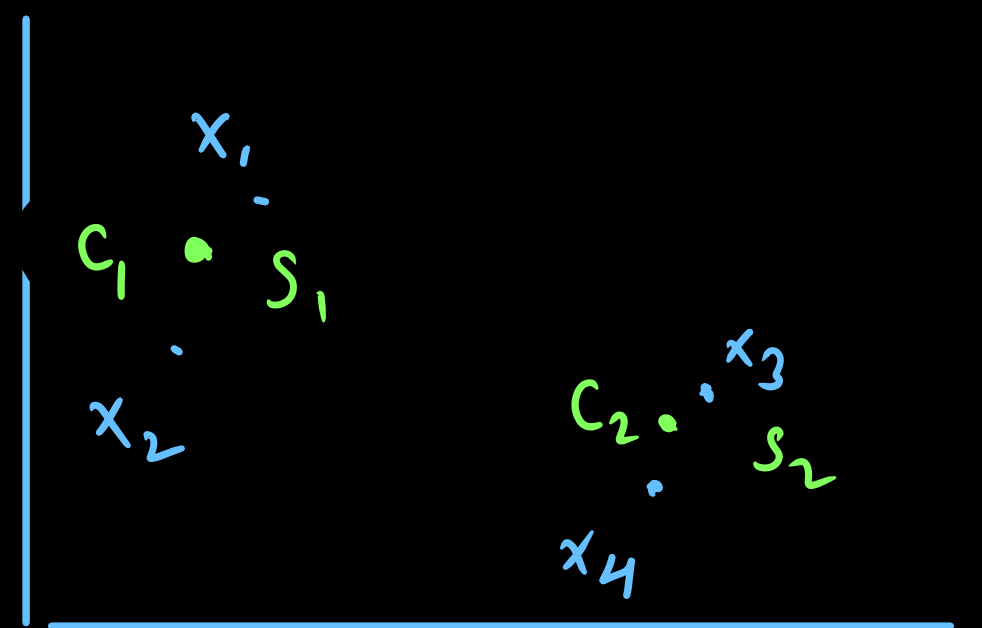
K-means

✓✓

$$Z = \begin{matrix} & S_1 & S_2 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \end{matrix}$$

GMM

$$Z = \begin{matrix} & S_1 & S_2 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.85 & 0.15 \\ 0.15 & 0.85 \\ 0.08 & 0.92 \end{bmatrix} \end{matrix}$$



$$z_{ij} = \begin{cases} 1 & x_i \in S_j \\ 0 & \text{otherwise} \end{cases}$$

$$z_{ij} = \text{Probab b/w } 0 \text{ \& } 1$$
$$\sum_{j=1}^K z_{ij} = 1$$

✓
$$L = \min_{C, Z} \|X - ZC^T\|_F^2$$

↓
→

Cluster Assignment Matrix Cluster centroid

GMM

S_1 S_2

$$Z = \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.85 & 0.15 \\ 0.15 & 0.95 \\ 0.08 & 0.92 \end{bmatrix}$$

Matrix
Decomposition

- MF
- PCA, SVD
- K-Means
- GMM

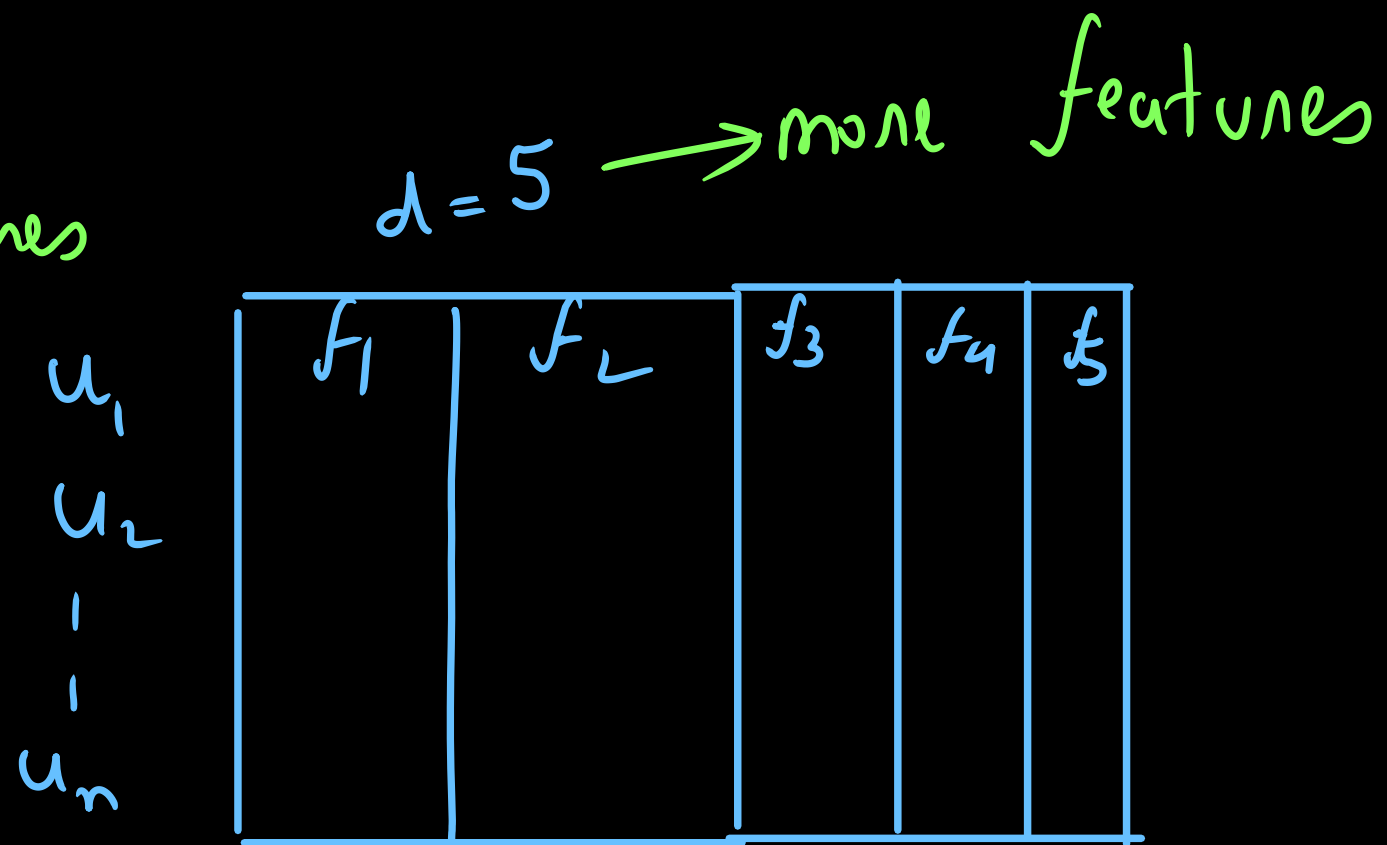
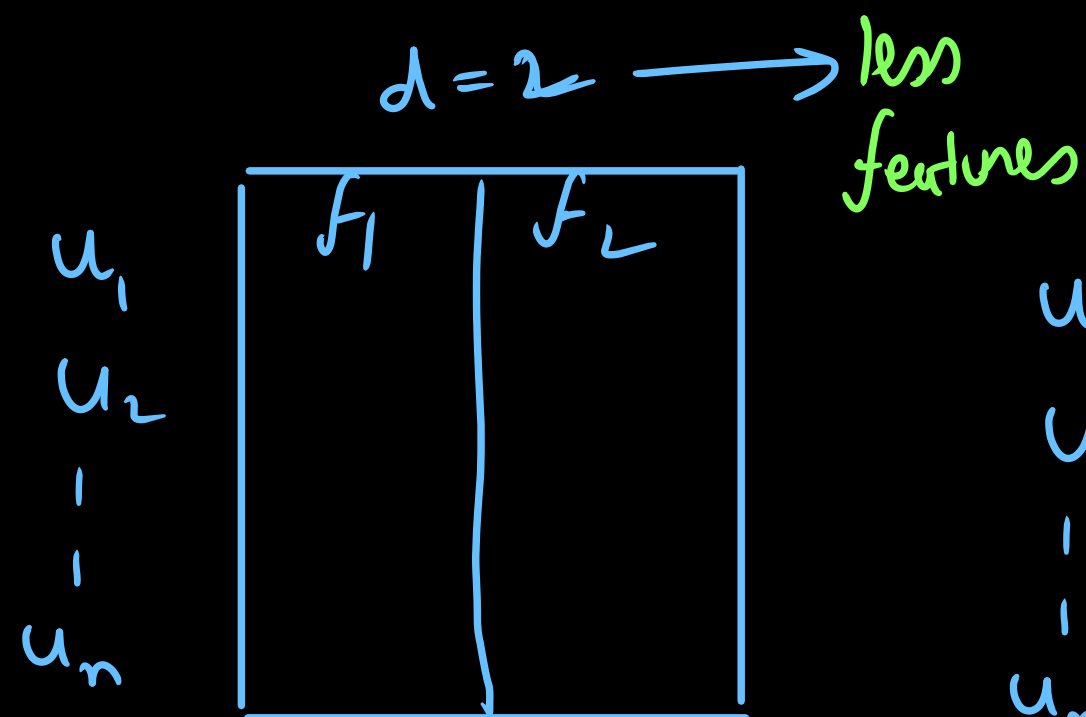
→ Break until 22:22 PM

→ Hyperparameter d

$$A_{n \times m} \approx \underbrace{B_{n \times d} C^T_{d \times m}}_{\tilde{A}}$$

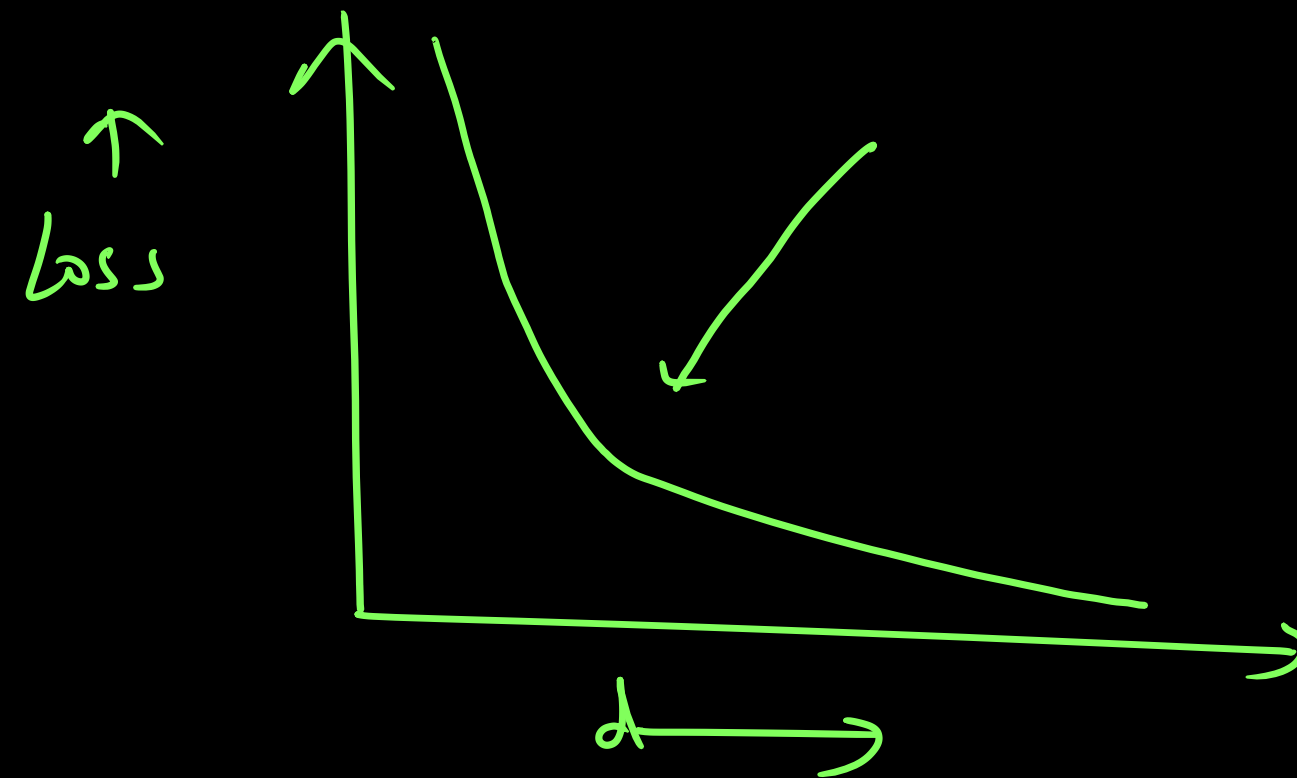
$$d < \min(m, n)$$

$$d \uparrow$$



① Train - Validation

② Elbow Method



NMF (Non - Negative MF)

$$A_{n \times m} \approx B_{n \times d} C_{d \times m}^T \quad \leftarrow \text{MF} \checkmark$$

$$A_{n \times m} \approx B_{n \times d} C_{d \times m}^T \quad \rightarrow \text{NMF}$$

$$B_{ij} \geq 0 \quad \forall i, j$$

$$C_{ij} \geq 0 \quad \forall i, j$$

Squid
Game



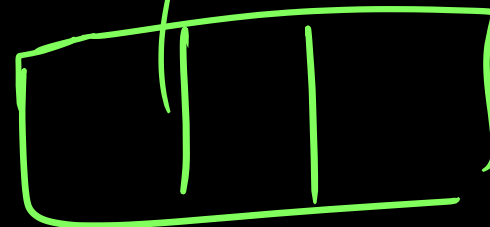
1-dim



π π π



1-dim



Kel ho
ho ho

1-dim

